

Water Chemistry Specification

Master Design Document: Batch-Sparge Centric Water Chemistry & Mash pH Calculator

1. Overview & Objectives

This document establishes the master architectural, functional, and user interface specifications for a minimalist, production-grade brewing water calculator tailored specifically for **single-stage batch sparge systems** (where Mash Volume V_1 and Sparge Volume V_2 may differ).

The design philosophy prioritizes a “minimum set of knobs” approach, abstracting complex stoichiometry and multivariable regression into a streamlined user experience while mathematically decoupling mash conversion chemistry from lautering water management. Furthermore, the calculator fully automates acid mitigation prescriptions, utilizes pre-ingested baseline profiles for tap water, consumes pre-supplied read-only grist category breakdowns, and explicitly incorporates **Acid Malt** into both the UI wireframe and the calculation pipeline.

2. User Interface Specification & Wireframe Layout

To maintain the minimalist philosophy while cleanly supporting advanced overrides, pre-supplied read-only data, and a batch-sparge workflow, the interface is structured as a single-page, responsive dashboard divided into visual blocks.

A. UI Wireframe Layout

```
+-----+
| BREW WATER & MASH CALCULATOR |
+-----+
| [1. BASE WATER & VOLUMES] |
| Source Water Profile: ( ) 100% R0 / Distilled (*) Tap Water |
| Mash Volume (V1): [ 5.5 ] gal Sparge Volume (V2): [ 6.5 ] gal |
+-----+
| [2. STYLE & GRAIN BILL (Pre-Supplied / Read-Only)] |
| Target Style: [ Hoppy / West Coast IPA ▼ ] |
| [+] Advanced Ion Ratio Override (Optional: Current 4.0 : 1) |
| | |
| Grist Breakdown (Total: 12.5 lbs) [Read-Only] |
| - Base Malts (Pils/Pale/2-Row): [ 10.5 ] lbs |
| | |
```

	- Crystal / Caramel Malts:	[1.5] lbs	
	- Roasted Malts:	[0.0] lbs	
	- Acid Malt:	[0.5] lbs	
+-----+			
	[3. ADJUSTMENT PREFERENCES] (Collapsible / Advanced)		
	- Salt Distribution:	[Mash-Centric (V1 only) ▼]	
	- Acid Medium:	[Lactic Acid (88%) ▼]	
	- Target Mash pH:	[5.4]	
+-----+			
	[LIVE PRESCRIPTION OUTPUT CARD] (Sticky / Bottom Panel)		
	--> MASH WATER ADDITIONS (V1): Gypsum 4.2g, CaCl2 1.5g, Acid 1.2mL		
	--> SPARGE WATER ADDITIONS (V2): Plain RO (No additions needed)		
	--> Projected Mash pH: 5.38 (Optimal / Green Indicator)		
+-----+			

B. Component Breakdown & UX Interaction Details

- Base Water & Volumes Card:** Features a segmented control between *100% RO / Distilled* and *Tap Water*. Selecting *Tap Water* automatically loads the pre-ingested, system-stored baseline tap water ionic array (Ca, Mg, Na, SO₄, Cl, Alkalinity), alongside numeric inputs for V_1 and V_2 .
- Style & Grain Bill Card:** Contains a macro-category searchable dropdown (*Soft Pilsner, Balanced Pale, Hoppy IPA, Malty Amber, Dark Stout*) that automatically injects default ion configurations. An expandable accordion allows advanced ratio overriding. The grist breakdown section renders the **pre-supplied grist category breakdown in a strict read-only format**, displaying all four buckets: Base Malts, Crystal / Caramel Malts, Roasted Malts, and Acid Malt.
- Preferences Card (Collapsible):** Manages salt distribution preferences (*Mash-Centric* vs. *Proportional*), liquid acid medium selection (*Lactic 88%* or *Phosphoric 10%*), and target mash pH setpoint. (*Note: When pre-supplied acid malt is present in the grist, its inherent acidification power is factored directly into the pipeline*).
- Live Prescription Output Card:** Reacts instantly on every state change. Includes color-coded health indicators (green for optimal 5.2–5.6 range, amber/red warning for tannin risks) and an explicit action split between V_1 and V_2 .

3. Required Input Data Set

The system ingests a compact dataset divided into pre-supplied/runtime parameters and internal system constants.

A. Parameters (Pre-Supplied / Recipe Provided)

- **Volume Parameters:** Mash Volume (V_1) and Sparge Volume (V_2).
- **Baseline Ion Array (ppm / mg/L):** Automatically bound to either zeroed vectors (for RO/Distilled) or the pre-ingested system baseline tap water profile containing Calcium (Ca^{2+}), Magnesium (Mg^{2+}), Sodium (Na^+), Sulfate (SO_4^{2-}), Chloride (Cl^-), and Total Alkalinity (as CaCO_3).
- **Grist Breakdown (Read-Only UI / System Ingested):** Mass values for the expanded set of four simplified malt categories:
 1. *Base Malts*
 2. *Crystal / Caramel Malts*
 3. *Roasted Malts*
 4. *Acid Malt*
- **Target Parameters:** Desired mash pH setpoint (default: 5.4) and optional user-mandated Sulfate:Chloride ratio (R).

B. System Constants & Lookup Matrices (Engine Provided)

- **Salt Dissolution Stoichiometry:** Fixed milligram-per-liter conversion values for 1 gram per gallon additions of Gypsum (CaSO_4), Calcium Chloride (CaCl_2), Epsom Salt (MgSO_4), Baking Soda (NaHCO_3), and Chalk (CaCO_3).
 - **Aggregate Malt Constants (Including Acid Malt):** Generalized defaults for the four malt categories:
 - *Base Malts:* $\text{pH}_{\text{DI}} \approx 5.75$, Buffer Index ≈ 45 mEq/kg/pH
 - *Crystal Malts:* $\text{pH}_{\text{DI}} \approx 5.00$, Buffer Index ≈ 30 mEq/kg/pH
 - *Roasted Malts:* $\text{pH}_{\text{DI}} \approx 4.70$, Buffer Index ≈ 15 mEq/kg/pH
 - *Acid Malt:* $\text{pH}_{\text{DI}} \approx 3.80$ (heavily loaded with lactic acid equivalent), Buffer Index ≈ 35 mEq/kg/pH
 - **Style Target Ion Dictionary:** Pre-mapped parameter arrays defining target Sulfate:Chloride defaults and calcium floors for macro-styles.
-

4. Calculation Pipeline (Two-Phase Batch Sparge Engine)

The computational engine processes inputs through a batch-sparge-aware pipeline separating conversion chemistry from lautering protection:

Phase A: Mash Chemistry & Treatment (V_1)**1. Dynamic Mineral Salt Optimization (Step 2):**

- *Ratio Transformation:* If a user mandates a custom Sulfate-to-Chloride ratio (R), the engine computes target individual ion bounds dynamically:

$$\text{Cl}_{\text{target}} = \frac{\text{Total Anion Target}}{R + 1}$$

$$\text{SO}_4^{\text{target}} = R \times \text{Cl}_{\text{target}}$$

- *Constrained Least-Squares:* Solves a non-negative matrix optimization ($x_n \geq 0$) to match the target ion vector against the active source water profile (RO or pre-ingested tap) while enforcing a strict minimum calcium floor (≥ 50 ppm) evaluated against V_1 .

2. Residual Alkalinity & Mash pH Prediction (Steps 3 & 4):

- Computes water Residual Alkalinity (RA) for V_1 via Kolbach's formula:

$$\text{RA}_1 = \text{Alkalinity}_1 - \left(\frac{\text{Calcium}_1}{1.4} + \frac{\text{Magnesium}_1}{1.7} \right)$$

- Balances V_1 RA strictly against the aggregate buffer index and pH_{DI} of the four grist buckets (incorporating the inherent acid contribution and buffering of the pre-supplied *Acid Malt* weight), properly normalized by total grist mass (m_{grist} in kg) and mash volume (V_1 in liters) to prevent unit scaling errors:

$$\text{Predicted Mash pH} = \overline{\text{pH}_{\text{DI}}} + \frac{\text{RA}_1 \cdot V_1 - \text{Liquid Acid Equivalent}}{\sum(m_{\text{grist}} \cdot \beta)}$$

3. Automated Liquid Acid Mitigation Prescription:

- Compares the projected raw mash pH (post-grist acid malt inclusion) against the Target Mash pH setpoint ($\text{pH}_{\text{target}}$). If additional mitigation is required, the engine automatically calculates the exact milliequivalent (mEq) deficit and translates it into an actionable automated liquid acid prescription (precise milliliters using standardized gram-equivalent normality constants for 88% Lactic or 10% Phosphoric) to drive equilibrium down to 5.2–5.6.

Phase B: Sparge Water Management (V_2)**1. Runnings Protection Analysis:**

- Evaluates V_2 starting alkalinity (from RO or the pre-ingested tap profile) to prevent high-pH extraction risks (tannin extraction occurs if combined runnings push kettle pH > 5.8).

2. Sparge Acidification Prescription:

- If the pre-ingested tap water profile is used for V_2 with non-zero alkalinity, prescribes a secondary, minor automated acid addition directly to the sparge water to neutralize its residual alkalinity (RA_2), ensuring safe lautering runnings (bypassed if using 100% RO/Distilled for V_2).

5. Output Prescription Interface

The system delivers a clean, split action list to the live output card for the brewer:

- **Mash Additions (V_1):** Exact salt weights and fully calculated supplementary liquid acid volume (accounting for the pre-supplied acid malt) targeted specifically to the strike water.
- **Sparge Additions (V_2):** Independent salt splitting allocation (if applicable) and protective sparge acidification volume.